

# Protocol city

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 1.3 Computer networks, connections and protocols — Paper 1, Computer systems                             |
| Lesson          | Protocol city (lesson 12 of the unit)                                                                             |
| Lesson ID       | nw-112                                                                                                            |
| Experience ID   | nw-112                                                                                                            |
| Plan ID         | PLAN-NW-L12                                                                                                       |
| Content version | b38bbd3c5f97                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=nw-112">https://revise360.co.uk/experience.html?id=nw-112</a> |

## Big question, outcomes and success criteria

**Big question** — What are standards and protocols?

### Prerequisites

The lesson before this one is nw-111. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: nw-101, nw-102, nw-103, nw-104, nw-105, nw-106, nw-107, nw-108-test, nw-109, nw-110.

### Assessable outcomes

- nw-112-o1** — understand the need for standards in computing
- nw-112-o2** — understand the seven common protocols and what they are used for

## Vocabulary, resources and preparation

**Key vocabulary** — Standard, Protocol

### Resources

|                   |                                               |
|-------------------|-----------------------------------------------|
| SLIDES            | presentations/NW_L12_ProtocolCity_Lesson.pptx |
| WORKSHEET         | worksheets/NW_L12_ProtocolCity_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=nw-112                     |

### Before the lesson

- Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
- Print the worksheet, one per pupil.
- Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
- Check headsets or devices are charged and on the school network.

### How pupils get in

- Go to `topics.html` on `revise360.co.uk`.
- Sign in with the **username and PIN from the login card**. The class is already set against the card.
- Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                              | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | Retrieve                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on NW_L12_ProtocolCity_Worksheet.docx is filled in.                    |
| 5-8   | Review the starter                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | Explain and model                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | Set up and settle                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find Standards.                                       | Everyone is in and on station 1.                                                       |
| 21-43 | Online: work the stations in order | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | Check understanding                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | Apply                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to d)                                                      |
| 56-60 | Exit and set up the next lesson    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. Standards

- Agreed rules and specifications so hardware and software from different makers work together.
- Examples: USB, Wi-Fi (IEEE 802.11), Ethernet and HTML.
- Without standards, devices couldn't talk to each other.

**Challenge on this wall:** What would happen if every phone maker used a different charging connector?

### 2. Protocols

- A protocol is a set of rules that governs how devices communicate on a network.
- It covers the format of data, addressing and how errors are handled.
- Different protocols do different jobs.

**Challenge on this wall:** What is the difference between a standard and a protocol?

### 3. Web protocols

- HTTP (Hypertext Transfer Protocol): a client-server way of requesting and delivering HTML files.
- HTTPS (Hypertext Transfer Protocol Secure): adds encryption and authentication.
- Look for the padlock in your browser.

**Challenge on this wall:** Why should a shopping website use HTTPS rather than HTTP?

#### 4. Email protocols

- SMTP (Simple Mail Transfer Protocol) sends email to a mail server.
- POP (Post Office Protocol) retrieves email from a mail server.
- IMAP (Internet Message Access Protocol) retrieves email and manages it on the server, so it syncs across devices.

**Challenge on this wall:** Why is IMAP better than POP if you read email on a phone and a laptop?

#### 5. FTP

- File Transfer Protocol: sends files between computers.
- Often used to upload web pages and their files to a web server for hosting.

**Challenge on this wall:** A web designer has finished a site. Which protocol might they use to put it on the web server?

#### 6. TCP/IP

- TCP (Transmission Control Protocol): splits data into packets and makes sure they all arrive, error-free and in order.
- IP (Internet Protocol): routes packets across a wide area network using IP addresses.
- They work together as TCP/IP.

**Challenge on this wall:** What would happen to a web page if some packets went missing and nothing checked?

### Retrieval and the check for understanding

**Starter** — List every acronym you already associate with the internet. Which ones do you think are rules for sending data, and which are something else?

On the board or the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

#### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (nw-112-s3-c2), so the wording a pupil sees matches the wording you say.

**What does HTTPS add to HTTP?**

**Correct:** Encryption and authentication of the data between client and server

#### What a pupil might choose instead

- Faster downloads
- Email attachments
- Bigger images

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** The S stands for Secure.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

### Guided to independent practice

| STAGE     | ACTIVITIES | WHAT IT IS                                                                                              |
|-----------|------------|---------------------------------------------------------------------------------------------------------|
| Recognise | 13         | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair. |

Every activity in this lesson is recognition: a pupil chooses between options and never produces anything of their own. Plan the independent work yourself - the exam practice on the worksheet is the obvious place - and do not read a full score here as evidence that a pupil can do it unaided.

## Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

## Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL**: pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND**: the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

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### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #     | STATION                         | ACTIVITIES | RECORD ON PAPER                       |
|-------|---------------------------------|------------|---------------------------------------|
| 1     | Standards                       | 2          | Standards are needed because...       |
| 2     | Protocols                       | 2          | A protocol is...                      |
| 3     | Web protocols                   | 2          | HTTP is used for... HTTPS adds...     |
| 4     | Email protocols                 | 2          | SMTP... POP... IMAP...                |
| 5     | FTP                             | 2          | FTP is used for...                    |
| 6     | TCP/IP                          | 2          | TCP makes sure... IP...               |
| final | Final challenge: protocol match | 1          | the answer to the station's challenge |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

### Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

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| ID        | QUESTION                                                                        | MARKS | WHERE                                                  |
|-----------|---------------------------------------------------------------------------------|-------|--------------------------------------------------------|
| nw-112-x1 | Explain why agreed standards matter when devices from different makers connect. | 3     | worksheet, Exam practice a); deck, Exam practice slide |
| nw-112-x2 | Name the protocol used to send email to a mail server.                          | 1     | worksheet, Exam practice b); deck, Exam practice slide |
| nw-112-x3 | Describe what HTTPS adds to HTTP and why it matters.                            | 3     | worksheet, Exam practice c); deck, Exam practice slide |
| nw-112-x4 | Explain how TCP and IP work together when a web page is requested.              | 4     | worksheet, Exam practice d); deck, Exam practice slide |

### Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/nw-112.json`, which is not published with the website.

### Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Why are standards needed in computing?** — To make devices more expensive; To stop people making new products
- **Which of these is a standard?** — A web browser's logo; A student's password
- **What is a network protocol?** — A type of cable; A network's password
- **Why must both devices use the same protocol?** — So they have the same MAC address; To use less electricity
- **What is HTTP used for?** — Sending email; Uploading files

### Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- Why standards are needed
- The common protocols and what they do
- The difference between standards and protocols

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is nw-113, Parcel depot. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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